CODEFEST 2026 / DOMAIN 02

HACKATHON BATTLE PLAN

Battle Plan

VIEWPILOT PRO

AI-Powered Job Readiness Agent Platform

A four-hour execution blueprint for building, integrating, and demoing an agent-powered interview readiness platform.

TEAM	TEAM SIZE
Mooncraft	2 builders
DEADLINE	TIMEBOX
Oct 24 · 12:00 AM IST	~4 hours

NAVIGATION / 00

Contents.

A quick route through the product concept, architecture, build plan, and live demo.

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- Keep the core experience working first; layer on polish only when the critical path is safe.

> **CAUTION**
>
> **4 HOURS.** Every minute counts. This plan is ruthlessly prioritized. **Dev A** = Frontend/UI specialist · **Dev B** = API/Logic specialist Both work in parallel, merge every 45 minutes.

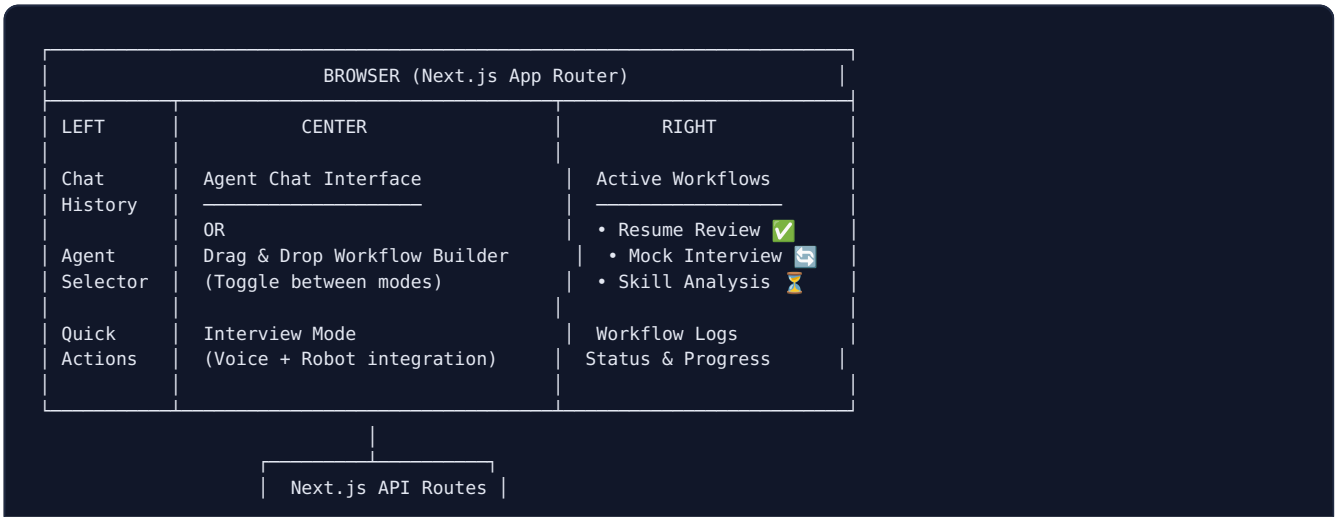
🎯 The Winning Concept

ViewPilot Pro — A job readiness platform where users interact with **specialized AI agents** through a ChatGPT-like interface, and can **visually build custom interview prep workflows** via drag-and-drop. The killer demo: a **robotic interviewer** powered by ElevenLabs voice + Gemini intelligence that conducts real interviews when the candidate can't be physically present.

Why This Wins

Judging Criteria	How We Nail It
Problem Fit	Directly addresses Domain 2's interview prep gap with an agent-based approach
Innovation	Drag-and-drop workflow builder for interview pipelines — nobody else will have this
Technical Depth	Multi-LLM (Gemini + OpenRouter), ElevenLabs TTS, ESP32-S3 robot, Next.js 16
UX/Design	Premium glassmorphic dark UI, scroll-driven 3D landing, Material Expressive
Demo Impact	Physical robot conducting an interview with AI voice = jaw-drop moment
Completeness	Landing, Auth, Dashboard, Workflow Builder, Interview, Settings — full product

🏗️ Architecture Overview



```
                    ┌──────────────────────┐
                    │ /api/agents/chat     │ ──► Gemini / OpenRouter
                    │ /api/agents/eval     │ ──► Gemini (evaluation)
                    │ /api/workflow/run    │ ──► Workflow engine
                    │ /api/interview/*     │ ──► Interview engine
                    │ /api/voice/tts       │ ──► ElevenLabs
                    │ /api/voice/stt       │ ──► Browser Web Speech
                    │ /api/robot/*         │ ──► ESP32-S3 (optional)
                    │ /api/auth/*          │ ──► NextAuth.js
                    └──────────────────────┘
```

📦 Tech Stack (Locked In)

Layer	Technology	Why
Framework	Next.js 16 (App Router)	SSR, API routes, RSC
Styling	Tailwind CSS 4 + Material Expressive UI (MUI)	Rapid premium UI
3D Landing	Three.js + @react-three/fiber + @react-three/drei	Scroll-driven 3D
Drag & Drop	@dnd-kit/core	Lightweight, accessible
AI (Primary)	Gemini 2.5 Flash via @google/genai	Fast, structured output
AI (Fallback)	OpenRouter (Claude/GPT fallback)	Reliability
Voice TTS	ElevenLabs API	Natural interviewer voice
Voice STT	Browser Web Speech API	Zero-cost, instant
Auth	NextAuth.js v5	Google OAuth + credentials
State	Zustand	Minimal, fast
Storage	localStorage + in-memory	No DB for hackathon
Robot	ESP32-S3 REST API (optional)	Physical demo wow-factor
Icons	Lucide React	Clean, consistent

🗺️ Page Map & Routes

```
/                   → 3D Scroll-Driven Landing Page
/login              → Login / Sign Up (NextAuth)
/dashboard          → Main Agent Interface (3-panel layout)
  ├── chat mode     → Talk to AI agents (Resume, Interview, Career)
  └── workflow mode → Drag-and-drop workflow builder
```

```
/interview          → Full interview session (voice + robot)
/settings           → API keys, preferences, voice settings
```

🎨 Design System — "Obsidian Glass"

Premium dark-mode glassmorphism. No AI clichés (no neural networks, no circuit boards, no glowing brains).

Color Palette

Token	Value	Usage
`--bg-primary`	`#0a0a0f`	Page background
`--bg-surface`	`rgba(255,255,255,0.03)`	Card/panel surfaces
`--bg-glass`	`rgba(255,255,255,0.06)`	Glassmorphic elements
`--border-glass`	`rgba(255,255,255,0.08)`	Subtle borders
`--accent-primary`	`#6C63FF`	Primary actions (indigo-violet)
`--accent-secondary`	`#00D9FF`	Secondary highlights (cyan)
`--accent-success`	`#22C55E`	Success states
`--accent-warning`	`#F59E0B`	Warning states
`--accent-error`	`#EF4444`	Error states
`--text-primary`	`#F0F0F5`	Headings, primary text
`--text-secondary`	`#9CA3AF`	Secondary text
`--text-muted`	`#4B5563`	Disabled/muted text

Typography

```
/* Google Fonts: Inter (body) + JetBrains Mono (code) + Outfit (headings) */
--font-heading: 'Outfit', sans-serif;
--font-body: 'Inter', sans-serif;
--font-mono: 'JetBrains Mono', monospace;
```

Glass Effect

```
.glass {
  background: rgba(255, 255, 255, 0.04);
  backdrop-filter: blur(24px) saturate(180%);
  border: 1px solid rgba(255, 255, 255, 0.06);
  border-radius: 16px;
```

```
  box-shadow: 0 8px 32px rgba(0, 0, 0, 0.4);
}
```

Micro-Animations

- Hover: `scale(1.02)` + border glow (`box-shadow` transition)
- Page transitions: `opacity` + `translateY(12px)` fade-up
- Workflow nodes: Spring-physics drag with `@dnd-kit`
- Chat messages: Slide-in from bottom with stagger
- Status indicators: Gentle pulse animation

🤖 The Five AI Agents

Each agent is a specialized persona with a system prompt, input/output schema, and evaluation rubric.

Agent	Icon	Role	API
Interview Coach	🎯	Conducts mock interviews, adapts questions, evaluates answers	Gemini
Resume Reviewer	📄	Analyzes resume text, scores sections, suggests improvements	Gemini
Skill Gap Analyzer	📊	Identifies missing skills for target role, creates learning path	OpenRouter
Career Advisor	🧭	Provides career guidance, role matching, industry insights	Gemini
Communication Trainer	🗣️	Evaluates answer clarity, suggests STAR formatting, tone coaching	Gemini

---

🔧 Workflow Builder — Drag & Drop Nodes

Pre-built nodes that users can chain into custom pipelines:

```
[Upload Resume] → [Resume Review] → [Skill Gap Analysis] → [Generate Questions]
                                                                    |
                                                            [Mock Interview]
                                                                    |
                                                           [Evaluate Answers]
                                                                    |
                                                        [Generate Report + Plan]
```

Node Types

Node	Inputs	Outputs
Upload Resume	File/text	Parsed resume data

Node	Inputs	Outputs
Set Target Role	Role, level, type	Config object
Resume Review	Resume data	Scores, suggestions
Skill Gap Analysis	Resume + target role	Gap list, learning path
Generate Questions	Role + level + gaps	Question set
Mock Interview	Questions + config	Transcript
Evaluate Answers	Transcript	Rubric scores, evidence
Generate Report	All above	PDF-ready report
Practice Plan	Evaluation	Micro-actions, STAR blueprint

Workflow Execution

When a workflow runs, each node executes sequentially. The right panel shows:

- Node name + status (⏳ Waiting → 🔄 Running → ✅ Done → ❌ Failed)
- Elapsed time per node
- Preview of each node's output
- Overall workflow progress bar

📞 Interview Mode — The Robot Demo

This is the **wow moment**. A candidate sits in front of the laptop (or robot), and the AI conducts a full voice interview.

Flow

```
1. User selects role, level, type, difficulty
2. System generates 5 role-specific questions (Gemini)
3. For each question:
   a. AI speaks the question (ElevenLabs TTS)
   b. Robot shows appropriate face expression (optional)
   c. Candidate answers via microphone (Web Speech API STT)
   d. AI evaluates in real-time, decides: follow-up or next question
4. Full evaluation report generated
5. Practice plan with STAR blueprints
```

Robot Integration (Optional)

```
Browser ──► /api/robot/expression ──► ESP32-S3 (face display)
Browser ──► /api/robot/status     ──► ESP32-S3 (health check)
Browser ──► /api/robot/gesture    ──► ESP32-S3 (head nod/tilt)
```

The robot adds **physical presence** — it shows facial expressions synced with the interview (thinking face while evaluating, nodding while listening, encouraging smile after a good answer).

📁 Project Structure

```
viewpilot-pro/
├── public/
│   ├── fonts/
│   └── models/                    # 3D assets for landing page
├── src/
│   ├── app/
│   │   ├── layout.tsx             # Root layout + providers
│   │   ├── page.tsx               # 3D Landing page
│   │   ├── login/
│   │   │   └── page.tsx           # Auth page
│   │   ├── dashboard/
│   │   │   └── page.tsx           # Main 3-panel agent interface
│   │   ├── interview/
│   │   │   └── page.tsx           # Voice interview session
│   │   ├── settings/
│   │   │   └── page.tsx           # Settings page
│   │   └── api/
│   │       ├── auth/[...nextauth]/
│   │       │   └── route.ts       # NextAuth config
│   │       ├── agents/
│   │       │   ├── chat/route.ts  # Agent chat endpoint
│   │       │   └── eval/route.ts  # Evaluation endpoint
│   │       ├── workflow/
│   │       │   └── run/route.ts   # Workflow execution
│   │       ├── interview/
│   │       │   └── route.ts       # Interview session
│   │       ├── voice/
│   │       │   └── tts/route.ts   # ElevenLabs proxy
│   │       └── robot/
│   │           └── route.ts       # Robot bridge
│   ├── components/
│   │   ├── landing/
│   │   │   ├── Hero3D.tsx         # Three.js hero scene
│   │   │   ├── FeatureSection.tsx # Scroll-triggered features
│   │   │   ├── WorkflowDemo.tsx   # Animated workflow preview
│   │   │   └── CTASection.tsx     # Final call-to-action
│   │   ├── dashboard/
│   │   │   ├── Sidebar.tsx        # Left: history + agent selector
│   │   │   ├── ChatPanel.tsx      # Center: chat interface
│   │   │   ├── WorkflowBuilder.tsx # Center: drag-and-drop builder
│   │   │   ├── WorkflowNode.tsx   # Individual draggable node
│   │   │   └── StatusPanel.tsx    # Right: running workflows
│   │   ├── interview/
│   │   │   ├── InterviewSetup.tsx # Role/level/type selector
│   │   │   ├── InterviewSession.tsx# Active interview with voice
│   │   │   └── InterviewReport.tsx # Post-interview evaluation
│   │   ├── auth/
│   │   │   └── AuthForm.tsx       # Login/signup form
│   │   ├── settings/
│   │   │   └── SettingsPanel.tsx  # Settings controls
│   │   └── ui/
│   │       ├── GlassCard.tsx      # Reusable glass card
│   │       ├── Button.tsx         # Styled button variants
│   │       ├── Badge.tsx          # Status badges
│   │       └── AnimatedText.tsx   # Text reveal animation
│   ├── lib/
│   │   ├── agents/
│   │   │   ├── prompts.ts         # System prompts for each agent
│   │   │   ├── interview-engine.ts # Core interview logic
│   │   │   └── evaluator.ts       # Answer evaluation + rubric
│   │   ├── ai/
│   │   │   ├── gemini.ts          # Gemini client
│   │   │   ├── openrouter.ts      # OpenRouter client
```

```
│   │   │   └── fallback.ts         # Local deterministic fallback
│   │   ├── voice/
│   │   │   ├── elevenlabs.ts       # ElevenLabs TTS client
│   │   │   └── speech.ts           # Web Speech API wrapper
│   │   ├── workflow/
│   │   │   ├── engine.ts           # Workflow execution engine
│   │   │   ├── nodes.ts            # Node type definitions
│   │   │   └── templates.ts        # Pre-built workflow templates
│   │   ├── robot/
│   │   │   └── client.ts           # ESP32-S3 HTTP client
│   │   └── store/
│   │       ├── chat-store.ts       # Zustand: chat state
│   │       ├── workflow-store.ts   # Zustand: workflow state
│   │       └── interview-store.ts  # Zustand: interview state
│   └── styles/
│       └── globals.css             # Design tokens + global styles
├── .env.example
├── .env.local
├── next.config.ts
├── tailwind.config.ts
├── tsconfig.json
└── package.json
```

⚡ API Endpoints

POST /api/agents/chat

```
// Request
{
  "agent": "interview-coach" | "resume-reviewer" | "skill-analyzer" | "career-advisor" | "communication-trainer",
  "messages": [{ "role": "user" | "assistant", "content": "..." }],
  "context": { "targetRole": "Frontend Engineer", "level": "Entry" }
}
// Response (streaming)
{
  "role": "assistant",
  "content": "...",
  "source": "gemini" | "openrouter" | "local",
  "metadata": { "tokens": 142, "latency_ms": 890 }
}
```

POST /api/workflow/run

```
// Request
{
  "workflow": {
    "id": "wf_123",
    "nodes": [
      { "type": "set-target-role", "config": { "role": "Frontend Engineer", "level": "Entry" } },
      { "type": "generate-questions", "config": { "count": 5, "type": "mixed" } },
      { "type": "mock-interview", "config": { "voice": true } }
    ],
    "edges": [["node_0", "node_1"], ["node_1", "node_2"]]
  }
}
// Response (SSE stream)
{ "nodeId": "node_0", "status": "running" }
{ "nodeId": "node_0", "status": "complete", "output": { ... } }
{ "nodeId": "node_1", "status": "running" }
...
```

POST /api/interview/session

```
// Request
{
  "action": "start" | "answer" | "next" | "evaluate",
  "sessionId": "sess_abc",
  "setup": { "role": "...", "level": "...", "type": "...", "difficulty": "..." },
  "answer": "My experience with React includes..."
}
// Response
{
  "question": "Tell me about a challenging project...",
  "questionNumber": 2,
  "totalQuestions": 5,
  "followUp": false,
  "evaluation": null | { "scores": {...}, "evidence": [...], "nextPractice": [...] }
}
```

POST /api/voice/tts

```
// Request
{ "text": "Tell me about your experience...", "voiceId": "interviewer_professional" }
// Response: audio/mpeg stream
```

GET /api/robot/status

```
// Response
{ "online": true, "battery": 85, "expression": "neutral", "ip": "192.168.1.42" }
```

⏱️ EXECUTION TIMELINE — 4 Hours

Phase 0: Setup (0:00 – 0:15) — BOTH

```
npx -y create-next-app@latest ./ --typescript --tailwind --eslint --app --src-dir --no-turbopack --import-alias "@/
*"
npm install @mui/material @emotion/react @emotion/styled
npm install zustand @dnd-kit/core @dnd-kit/sortable @dnd-kit/utilities
npm install three @react-three/fiber @react-three/drei
npm install @google/genai
npm install next-auth@beta
npm install lucide-react
npm install framer-motion
```

- Set up `.env.local` with API keys
- Create base `globals.css` with design tokens
- Create `layout.tsx` with fonts + providers

Phase 1: Core Foundation (0:15 – 1:15) — PARALLEL

Dev A: Landing Page + Auth UI

Time	Task
0:15 – 0:45	**3D Landing Page** — Hero with Three.js scene (floating geometric shapes — dodecahedrons, torus knots, spheres — with glassmorphic panels overlaid). Scroll-driven animations with Framer Motion. Feature sections, workflow preview animation, CTA.
0:45 – 1:00	**Login/Signup Page** — Glass card with email/password form, Google OAuth button, animated background.
1:00 – 1:15	**Global Layout** — Navigation bar, page transitions, responsive breakpoints.

Dev B: API Layer + AI Integration

Time	Task
0:15 – 0:35	**Gemini client** (`lib/ai/gemini.ts`) — Structured output, streaming, system prompts for all 5 agents.
0:35 – 0:50	**OpenRouter client** (`lib/ai/openrouter.ts`) — Fallback LLM, same interface.
0:50 – 1:00	**ElevenLabs client** (`lib/voice/elevenlabs.ts`) — TTS proxy endpoint.
1:00 – 1:15	**Agent chat API** (`/api/agents/chat`) — Streaming endpoint, agent routing, context handling.

Phase 2: Dashboard (1:15 – 2:30) — PARALLEL

Dev A: Dashboard UI (The 3-Panel Layout)

Time	Task
1:15 – 1:35	**Left Sidebar** — Agent selector (5 agents with icons), chat history list, collapsible panel.
1:35 – 2:00	**Center: Chat Interface** — Message bubbles (glass style), typing indicator, markdown rendering, input bar with voice toggle.
2:00 – 2:15	**Center: Workflow Builder** — Toggle between Chat/Workflow mode. Drag-and-drop canvas with `@dnd-kit`. Node palette + drop zone.
2:15 – 2:30	**Right Panel** — Active workflows list with status badges (⏳🔄✅❌), expand to see node-level progress.

Dev B: Workflow Engine + Interview API

Time	Task
1:15 – 1:40	**Workflow engine** (`lib/workflow/engine.ts`) — Sequential node executor, SSE streaming status updates.

Time	Task
1:40 – 2:00	**Node definitions** (`lib/workflow/nodes.ts`) — All 9 node types with execute functions.
2:00 – 2:15	**Workflow API** (`/api/workflow/run`) — SSE endpoint, workflow validation, execution.
2:15 – 2:30	**Interview API** (`/api/interview/session`) — 5-question session, follow-up logic, evaluation.

Phase 3: Interview Page + Polish (2:30 – 3:30) — PARALLEL

Dev A: Interview UI + Settings

Time	Task
2:30 – 2:50	**Interview Setup** — Role/level/type/difficulty selectors with Material Expressive styling.
2:50 – 3:15	**Interview Session** — Live question display, voice recording indicator, real-time transcript, timer, robot status indicator. Question progress bar.
3:15 – 3:30	**Settings Page** — API key inputs, voice preferences, robot URL config, theme toggle placeholder.

Dev B: Voice Integration + Robot Bridge + Evaluation

Time	Task
2:30 – 2:50	**Web Speech integration** — STT wrapper, real-time transcription in interview.
2:50 – 3:10	**ElevenLabs TTS integration** — Speak questions aloud, audio playback management.
3:10 – 3:20	**Robot bridge** (`/api/robot/*`) — Expression sync, status polling.
3:20 – 3:30	**Evaluation engine** — 4-dimension rubric scoring, evidence extraction, practice plan generation.

Phase 4: Final Integration + Demo Prep (3:30 – 4:00) — BOTH

Time	Task
3:30 – 3:40	**Integration testing** — Full workflow: Landing → Login → Dashboard → Chat → Workflow → Interview → Report.
3:40 – 3:50	**Polish** — Fix layout breaks, add loading states, error boundaries, smooth transitions.
3:50 – 3:55	**Environment check** — Verify all API keys work, test with/without robot.
3:55 – 4:00	**Git push** — Clean commit, update README, verify build.

🎤 Demo Script (3 minutes)

Minute 1: The Problem + Landing Page

> "Students never get real interview practice with adaptive feedback. They use static question banks and get no structured improvement path."
>
> *[Show the landing page — smooth 3D scroll, feature reveals, architecture diagram]*

Minute 2: The Platform

> "ViewPilot Pro gives you 5 specialized AI agents — chat with them like ChatGPT, or build custom interview prep workflows with drag-and-drop."
>
> *[Show dashboard — chat with Interview Coach, switch to Workflow Builder, drag nodes, run a workflow, show right panel status]*

Minute 3: The Robot Interview (Wow Moment)

> "And for candidates who can't physically attend — our AI robot conducts the interview for them. Real voice. Real evaluation. Real feedback."
>
> *[Start interview — AI speaks question via ElevenLabs, candidate answers, robot shows expressions, evaluation report appears]*
>
> "Four rubric scores. Evidence from your own words. A practice plan. All in 30 seconds."

🛡 Risk Mitigation

Risk	Mitigation
API rate limits	Local fallback engine for all agents — app works with zero API keys
ElevenLabs quota	Fall back to browser `speechSynthesis` API
Robot offline	Robot is always optional — interview works without it
3D performance	Lightweight geometry only (no heavy models), `Suspense` fallback
Time crunch	If behind: skip Settings page, simplify workflow builder to pre-built templates only

🔑 Environment Variables

```
# .env.local
GEMINI_API_KEY=your_gemini_key
OPENROUTER_API_KEY=your_openrouter_key
ELEVENLABS_API_KEY=your_elevenlabs_key
ELEVENLABS_VOICE_ID=your_voice_id

# Optional
ROBOT_BASE_URL=http://192.168.1.42
NEXTAUTH_SECRET=your_random_secret
NEXTAUTH_URL=http://localhost:3000
GOOGLE_CLIENT_ID=your_google_oauth_client_id
GOOGLE_CLIENT_SECRET=your_google_oauth_client_secret
```

📋 Priority Tiers (If Time Runs Short)

🔴 MUST HAVE (First 2.5 hours)

- ⊡ Landing page (even simplified — 2D with animations is fine if 3D takes too long)
- ⊡ Login page (working auth form, can skip OAuth)
- ⊡ Dashboard with chat interface (at least 1 working agent)
- ⊡ Interview page with text input (basic flow)
- ⊡ API routes for chat + interview

🟡 SHOULD HAVE (Next 1 hour)

- ☐ Workflow builder with drag-and-drop
- ☐ Right panel with workflow status
- ☐ ElevenLabs voice in interview
- ☐ Multiple working agents
- ☐ Evaluation report with rubric

🟢 NICE TO HAVE (Final 30 min)

- ☐ 3D elements on landing page
- ☐ Robot integration
- ☐ Settings page
- ☐ Pre-built workflow templates
- ☐ Smooth page transitions
- ☐ Print/export report

IMPORTANT

Key Principle: Every feature should work without external APIs. Build the local fallback first, then layer AI on top. The judges should never see a loading spinner or error screen.

Ready to Execute?

Click **Proceed** below and I will start scaffolding the entire project — Next.js app, all pages, components, API routes, design system — in the optimal build order for 2 developers working in parallel.